

ACRME Repo Consistency Audit — 17 September 2026

Scope. Full sweep of the ACRME documentation set (Requirements/ , Architecture/adr/ , Design/ , Reference-Material/ , POC-Test-Scripts/) checked against (a) the requirements baseline and (b) the canonical calculation formulas & constants. Archive/ excluded (historical). Auto-generated .docx / .pdf twins excluded (they regenerate from their .md).

Authority sources used for the audit

- Requirements: Requirements/acrme_requirements_baseline_v2_4.md — **v2.4 (amended), 11 September 2026** (the newest baseline in the repo).
- Formulas & constants: Reference-Material/reference/acrme_calculation_logic_reference.md .

Bottom line. The repo is internally consistent. The audit found **two** issues, both now **RESOLVED**: (1) the uploaded “source of truth” baseline was stale vs. the repo baseline — now synchronized to the amended 11-Sep version; (2) one internal typo (5-min vs. 6-min reconciliation interval) — corrected. All documents now reconcile to the authoritative baseline v2.4 (amended), 11 September 2026.

Finding 1 — HIGH — The uploaded “source of truth” baseline was STALE vs. the repo baseline [RESOLVED]

The standing instruction points at the uploaded file as the single source of truth, but it is an **older revision** than the baseline committed in the repo, and **every other document in the repo reconciles to the newer repo baseline, not the uploaded one.**

	Uploaded file	Repo baseline
Path	/home/ubuntu/Uploads/Azure Capacity & Quota Management- Consolidated Requirements Baseline.md	Requirements/acrme_requirements_baseline_v2_4.md
Version / date	v2.4, 7 September 2026	v2.4 (amended), 11 September 2026
Middle East DR	DR_NOT_OFFERED (“DR is unlikely to be offered there”)	Cross-geo DR to Europe (Prod+CVAl co-located in a weighted-selected ME region; DR weighted-selected in a Europe Standard region)
DEC-001	Pending / open	RESOLVED
Extra normative IDs	—	Adds DR-020, PLC-010b, A-ME1

Impact. All downstream docs — ADRs, FDD, TDD, calc-logic reference, POC workbook, epics/stories, complete requirements reference — already implement the **amended (11-Sep)** Middle East cross-geo position with DEC-001 RESOLVED. Examples:

- Requirements/acrme_complete_requirements_reference.md:103,302 — cross-geo DR into Europe (DR-020, PLC-010b, DEC-001 RESOLVED).
- POC-Test-Scripts/acrme_poc_workbook_v2_4.md:19,144,2167 — POC-PLC-010b positive cross-geo case; supersedes DR_NOT_OFFERED .
- Reference-Material/backlog/acrme_epics_stories_tasks.md:593,697 — ACRME-S0706 Middle East cross-geo DR (DEC-001 RESOLVED).

So the repo is coherent **against the amended repo baseline**; the divergence is only against the **uploaded** file the standing instruction names.

Resolution (confirmed by user): The amended 11-Sep baseline is authoritative. The uploaded file at /home/ubuntu/Uploads/Azure Capacity & Quota Management- Consolidated Requirements Baseline.md has been **synchronized** with the repo baseline — both now carry **v2.4 (amended), 11 September 2026** with Middle East cross-geo DR, DEC-001 RESOLVED, DR-020, PLC-010b, A-ME1. The standing instruction source of truth and the repo are now aligned.

Finding 2 — LOW — Reconciliation interval typo (FIXED in this change)

Design/acrme_uml_class_diagrams_summary.md:64 stated the reconciliation loop target interval as **5 minutes**. Every other source says **6 minutes (360 s), configurable**:

- Baseline **CAP-006**;
- Design/acrme_technical_design_document.md:135 — “6-minute target interval (configurable, CAP-006)”;
- Reference-Material/reference/acrme_calculation_logic_reference.md:610 .

Fix applied: line 64 now reads “6-min target interval (configurable, CAP-006)”.

Finding 3 — MEDIUM — Fixed DR ratio still live in Hard Constraints / ADR-007 (CONVERTED to max-not-sum in this change)

While the Calculation Logic Reference already sized DR with **max-not-sum** (A.6 / DR-017 / Scenario 17) and marked Scenario 15 superseded, several docs still **actively computed HC-6 / HC-7 with the fixed $dr_ratio_max = 0.40$** — the exact model the calc ref says must NOT be used. This was an internal inconsistency, now resolved by retiring the fixed dr_ratio_* model everywhere it was still a live formula and converting to max-not-sum plus the configurable DR bootstrap target (DR-007).

Files converted:

- Reference-Material/reference/acrme_hard_constraints_reference.md — HC-6 (per-customer DR demand $\rightarrow dr_bootstrap_qty$, DR-007), HC-7 DR floor ($DR_Floor_vCPU(R) = Destination_DR_Requirement(R) \times vCPU$, $Destination_DR_Requirement(R) = MAX(source\ portions)$), POC example reframed, policy JSON + constants table (dr_ratio_* marked **✗ Retired**).
- Reference-Material/reference/acrme_calculation_logic_reference.md — fixed internal inconsist-

ency in `coverage_ratio` denominator (was $\sum \text{potential_dr_demand}$, now

`Destination_DR_Requirement = MAX(...)`, matching line ~465); Scenario 15 reframed **Retired (historical rationale only)**; clarified the C-11 SUM override sums actual source portions and does **not** re-introduce `dr_ratio`; summary/disambiguation tables → Retired.

- `Architecture/adr/acrme_adr_007_crg_deployment_model.md` — Appendix B HC-6/HC-7 formulas converted to bootstrap + max-not-sum.
- `Requirements/acrme_complete_requirements_reference.md` — FR-5.6 traceability note updated to max-not-sum floor + configurable bootstrap.
- `Design/acrme_technical_design_document.md`, `Reference-Material/reference/AC-RME_Scoring_Weights_Explained.md`, `Reference-Material/reference/acrme_plain_english_walkthrough.md`, and calc ref `PS_DR` — the scoring δ term's normalization constant renamed `dr_ratio_target` → `dr_coverage_target` and annotated as a **scoring-only** reference (its `coverage_ratio` uses the max-not-sum denominator), NOT the retired sizing ratio.
- `POC-Test-Scripts/acrme_poc_workbook_v2_4.md` — assumption A-09 ("30-40% baseline is sufficient") marked **RETIRED**.

Preserved (historical rationale, intentionally not removed): the \$1.5M-\$5M/year idle-cost justification (Scenario 15, ADR-002/003/005, baseline Appendix D), ENV-005 "must NOT default to fixed 30-40%", and C-1's ratio-evolution history. **Out of scope:** `Archive/` documents (historical snapshots) retain the old fixed-ratio formulas by design.

No conversion needed (already correct): FDD §4.5, ADR-002/003/005, comprehensive walk-through, code glossary C-1, and the baseline itself — all already state max-not-sum / configurable bootstrap and reject the fixed ratio.

Items reviewed and confirmed **CONSISTENT (no action)**

- **Formulas A.1-A.9** (baseline Appendix A) match the calc-logic reference — including A.6 DR destination sizing = **MAX not SUM** over sources (DR-017) and A.9 even-zone share = $1/\text{zone_count}$.
- **Scoring weights** $\alpha=0.30$, $\beta=0.20$, $\gamma=0.25$, $\delta=0.15$, $\epsilon=0.10$ (sum 1.0) — consistent wherever cited (marked `[Assumed]` in the calc ref, which is correct; the baseline does not fix them numerically).
- **Growth buffers** `prod/nonprod = 0.20`, emergency transfer = 0.30; **auto-increase thresholds** `dr=0.35`, `prod=0.20`, `nonprod=0.20`; debounce 30 min — consistent.
- **Region model** (US 3-region; EU/AU/APAC 2-region co-located; Middle East cross-geo) — consistent across ADRs, FDD, TDD, POC, backlog.
- **ADRs** — no inconsistencies. ADR-007 is intentionally `PROPOSED` (pending POC-001/006, DEP-001); its §15 note that CAP-013 wording says "up to ~100" while Azure's hard limit is exactly 100 is a documented wording nit, not a contradiction.

Known open design gaps (not inconsistencies — noted for awareness)

- **total_customers denominator** in the γ (fairness) term is not defined in the baseline. Already captured as an OPEN ISSUE in `Reference-Material/reference/acrme_calculation_logic_comprehensive_walkthrough.md` (prior commit) — left as-is.

- **PS_NonProd design-of-record** intentionally combines α and δ signal (0.45); flagged in the PRR with a corrected pilot variant — intentional, not an error.
- **Scenario 15** (fixed `dr_ratio`) is now **RETIRED** and every live HC-6/HC-7 formula has been converted to max-not-sum (see Finding 3). Historical rationale retained by design.